

Update on the Pennsylvania Waste Tire Recycling Program

July 30, 2004

Commonwealth of Pennsylvania
Edward G. Rendell, Governor

Department of Environmental Protection
Kathleen A. McGinty, Secretary
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Meeting our Challenge

**Kathleen A. McGinty, Secretary
Department of Environmental Protection**

It is my pleasure to present this report to the General Assembly

The purpose of this report is two-fold. First, it responds to the Legislature's requirement that state agencies give due consideration to the use of waste tires in construction and engineering applications. It also provides a progress report on the larger issue of waste tires and abandoned tire piles.

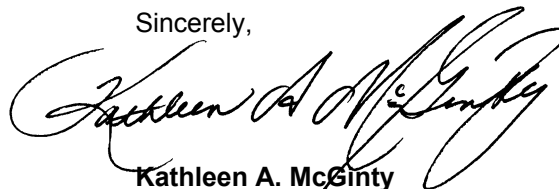
In 1996, the Pennsylvania Department of Environmental Protection (DEP) estimated that approximately 36 million tires lay abandoned in stockpiles across the Commonwealth. Not only were these piles a visual scar on Pennsylvania's landscape, they were causing serious environmental, health and safety problems, including hazards associated with tire fires and heavy mosquito infestations. Since then, the Commonwealth has remediated nearly 24 million waste tires. Our success is based on a combination of adequate funding for the cleanup of abandoned tire piles and enforcement and compliance actions at commercial waste tire piles.

Today, approximately 12.5 million waste tires remain in approximately 70 stockpiles across Pennsylvania. More than 8.5 million waste tires (68 percent) are contained in piles with no viable responsible party. The largest are the Max and Martha Starr pile in Columbia County, which contains an estimated six to ten million waste tires, and Natrl Windgap in Northampton County, with an estimated 1.2 million waste tires.

The remaining 4 million waste tires are located at commercial sites where the Department is taking enforcement and compliance actions to reduce the number of tires and the risks from fires and mosquitoes. The largest commercial pile is Mahantango Enterprises in Juniata County with an estimated 1.5 million tires.

While we have risen to the challenge of identifying, reducing and remediating the waste tire stockpiles in the Commonwealth; implementing the waste tire hauler program and encouraging the reuse of waste tires, more needs to be done. In particular, continued funding is needed to cleanup the remaining abandoned piles, and amendments to the Waste Tire Recycling Act are necessary to improve our ability to place liens on remediated piles and prevent new waste tire piles. I look forward to your continued support in this endeavor.

Sincerely,

A handwritten signature in black ink, appearing to read 'Kathleen A. McGinty', written over a horizontal line.

**Kathleen A. McGinty
Secretary**

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Summary

Commonwealth Use of Waste Tires

By July 30, 2004, several Pennsylvania state agencies were required to give due consideration to the use of waste tires in all appropriate construction and engineering activities which are paid with public funds. As appropriate, the majority of listed Departments have utilized waste tire products.

The actual use of waste tires in construction and engineering activities was limited by:

- The small number of acceptable uses for waste tires in construction and engineering activities;
- The small number of construction and engineering activities where waste tires could be used; and
- The lack of knowledge about the type of construction and engineering projects that could use waste tires.

While there is a developing market for waste tire products, it is still growing and not entirely sustainable or cost-competitive with other materials used for the same projects. The majority of these uses do not involve construction and engineering activities. At present, tire-derived fuel (TDF) is the primary consumer of waste tires.

Waste Tire Hauler Authorization Program

The Department has implemented the Waste Tire Hauler Authorization program as required under the Act 111 amendments to Act 190. To date, the Department has authorized approximately 775 individuals, companies and agencies that collect and haul waste tires to processors.

Waste Tire Pile Cleanup

Through the implementation of Act 190 and other related waste tire programs, Pennsylvania has removed approximately 23.8 million waste tires from known waste tire piles. As a result, we have restored hundreds of acres of land to productive use. Pennsylvania's success is attributed to the implementation of stringent compliance initiatives and remediation grant programs that foster recycling and waste prevention.

Today, approximately 12.5 million waste tires remain in approximately 70 stockpiles across Pennsylvania, with more than 8.5 million waste tires (68 percent) contained in piles with no viable responsible party. The remaining 4 million tires are located at commercial sites where the Department is taking enforcement and compliance actions to reduce the number of tires, and the risks from fires and mosquitoes.

Commonwealth Recycling and Use of Waste Tires (Section 113 of Act 111)

Commonwealth Use of Waste Tires in Construction and Engineering Activities

Pennsylvania Act 111 of 2002, Section 6029.113 (a), states:

Use of waste tires by Commonwealth agencies. — By July 30, 2004, the Department of Conservation and Natural Resources, the Department of Corrections, the Department of Education, the Department of Environmental Protection, the Department of General Services, the Department of Transportation, the State System of Higher Education and the State-related universities shall, to the maximum extent practicable and feasible, give due consideration to the use of waste tires in all appropriate construction and engineering activities which are paid with public funds.

As appropriate, the majority of listed Departments have given due consideration to the use of waste tire products in construction and engineering activities. The actual use of waste tires in construction and engineering activities was limited by:

- The small number of acceptable uses for waste tires in construction and engineering activities;
- The small number of construction and engineering activities where waste tires could be used; and
- The lack of knowledge about the type of construction and engineering projects that could use waste tires.

While there is a developing market for waste tire products, it is still growing and not entirely sustainable or cost-competitive with other materials used for the same projects. The majority of these uses do not involve construction and engineering activities. At present, tire-derived fuel is the primary consumer of waste tires and finds more use in the private sector than it does in state agency projects.

Examples of construction and engineering projects undertaken by Commonwealth include:

- Corrections has been studying the use of TDF to supplement normal fuel sources, but no results of this study have been released.
- DEP approved the use of two-inch-minus tire chips as an alternate coarse aggregate for drainage material in on-lot septic systems. The chips replace crushed stone, normally used for drainage media.
- DEP and the Department of Transportation (PENNDOT) are collaborating on the Tarrtown Bridge project near Kittanning, Pennsylvania, that will utilize approximately 750,000 waste tires originating from waste tire cleanups and tire amnesty days in Armstrong and Indiana counties. The waste tires are shredded and used to construct two lightweight embankments for bridge approaches.
- The Department of Transportation has evaluated crumb rubber in hot-mix asphalt. The results have ranged from fair to satisfactory, with poor performance noted in open-graded asphalt surface course, due primarily to Pennsylvania's significant use of anti-skid material, which fills the voids in the mix and causes severe freeze/thaw damage. In all recent cases, costs were seven to 60 percent higher than the cost of standard hot-mix asphalt.
- PENNDOT used approximately 74,000 pounds of crumb rubber in a research project to evaluate Tyrsolv crumb rubber asphalt modifier. Costs ranged from seven to 23 percent higher than standard hot-mix asphalt. Although the Tyrsolv modifier did perform well in rut resistance, it showed no significant increase in performance over standard hot-mix asphalt.

The following is a detailed report describing the actions these agencies have taken in the previous two years to implement Section 113 Commonwealth Recycling and Use of as required by Pennsylvania Act 111 of 2002, Section 6029.113 (b)

Department of Conservation and Natural Resources

The Department of Conservation and Natural Resources is one of the state agencies that has the potential for significant use of waste tires in construction and engineering projects, particularly rubber chips from shredded waste tires. In the past, they have used waste tires in erosion control and have used materials derived from waste tires in playground matting and walking trails. They have also studied the use of tire chips for turf durability. Other potential uses for tire-derived products include groundwater drain aggregate, parking lot surfacing material,

additional athletic field surfaces, boat docks, and rubberized lumber for use in walkways, benches, posts, and fencing. Additionally, re-treaded waste tires can be used for Department vehicles.

Department of Corrections

The Department of Corrections has less opportunity to directly use waste tire-derived products as most of their facilities construction and engineering activities are carried out by the Department of General Services. Examples of DGS managed construction and engineering activities which should give due consideration to the use of waste tires, include: athletic surfaces, parking lots and walking surfaces. However, there are still possible uses within the Department of Corrections. As mentioned before, the use of TDF has been under study for use in supplementing facility fuel sources. Also, the use of rubberized surfaces in athletic facilities and walking surfaces would be a viable use in the department's facilities.

Department of Education

The Department of Education has less opportunity to directly incorporate waste tire-derived materials into construction and engineering projects than other departments because public school construction and engineering activities in the Commonwealth are handled at the district level. The Department of Education does not engage in any facility construction activities of its own. However, the Department does review and approve all plans and specifications for public school construction or repair plans at a value greater than \$15,000 in second, third, and fourth class school districts. In the course of review and approval, the Department of Education should ensure that due consideration was given to the use of waste-tire derived products when appropriate. The most likely tire-derived uses would include athletic surfaces, parking lots, playgrounds, and walking surfaces.

Department of Environmental Protection

The Department of Environmental Protection's (DEP) Bureau of Water Supply and Waste Water Management approved the use of two-inch-minus tire chips as an alternate coarse aggregate for drainage material in on-lot septic systems. The chips replace crushed stone, normally used for drainage media.

In 2000, DEP approved the use of a two-inch-minus tire chip as an alternative for gravel in the construction and installation of on-lot sewage disposal systems. The tire chips, which facilitate drainage just as conventional gravel does, have been found to be lighter and easier to use than conventional materials. It takes approximately 1,831 waste tires, or the equivalent of 36,630 pounds of rubber, to make enough tire chips to construct a septic system with an average size of 990 sq. ft.

In 2000, DEP issued a contract to process and eliminate waste tire piles with an on-site mobile tire shredder. The shredder, which was transported to waste tire piles across the state, processed waste tires into tire chips. The chips were transported directly to end users and used in a variety of applications, including landfill construction, other civil engineering uses and as tire derived fuel. The shredder contract expired on June 30, 2004.

DEP and the Department of Transportation (PENNDOT) are collaborating on the Tarrtown Bridge project near Kittanning, Pennsylvania, that will utilize approximately 750,000 waste tires originating from waste tire cleanups and tire amnesty days in Armstrong and Indiana counties. The waste tires will be shredded and used to construct two lightweight embankments for bridge approaches. This is the first project of its kind for the Commonwealth. The waste tires and instrumentation for the embankment fill are being provided by DEP. PENNDOT is providing the engineering design, project management and detailed inspections. It is hoped that the data obtained from this project will lead to additional uses of shredded tires as lightweight fill on other highway projects. There are several other PENNDOT districts that are planning to implement similar projects in the near future.

DEP has awarded grants to demonstrate the uses of waste tires and crumb rubber in applications such as playground equipment and safety surfaces, park trails, school tracks and stadiums, sports mats and top dressings for turfgrass. Top dressing — normally a mixture of sand and organic material — is applied to turfgrass to cushion it from the wear and tear of foot traffic. DEP has worked cooperatively with the Department of Conservation and Natural Resources and PENNDOT to develop many of the projects.

Department of General Services

The Department of General Services (DGS) has the potential to use a significant amount of waste tires and tire-derived products in both engineering and construction activities and general purchasing. As the primary agency responsible for facilities construction in the Commonwealth, DGS has the ability to make certain that waste tire-derived products are used whenever feasible. These uses include flooring and carpet padding, parking lots, parking lot bumpers, rubberized lumber, and rubberized surface coatings.

DGS also handles the Commonwealth's tire recycling contract, scrap tire removal and has contracts for the re-treading and repairing of tires. Additionally, DGS purchases traffic cones and delineation devices that can be produced from crumb rubber derived from waste tires.

Department of Transportation

PENNDOT has evaluated crumb rubber in hot-mix asphalt. The results have ranged from fair to satisfactory, with poor performance noted in open-graded asphalt surface course, due primarily to Pennsylvania's significant use of anti-skid material, which fills the voids in the mix and causes severe freeze/thaw damage. In all recent cases, initial costs were seven to 60 percent higher than the cost of standard hot-mix asphalt. Use of crumb rubber is permitted in bituminous concrete base courses (BCBC) at one percent of the total mixture weight, but few, if any, hot-mix asphalt producers employ this provision, citing increased costs and the state's competitive low-bid practice. Past experiments with crumb rubber in hot-mix asphalt reduced PENNDOT's buying power without an associated increase in performance of the hot-mix asphalt pavement; however, recent advances in chemical additives, chemically treated crumb rubber, and open-graded pavement mixtures show some potential for obtaining better or increased performance. PENNDOT will continue to evaluate these advances for their applicability in Pennsylvania and balance environmental, engineering, and economic considerations for waste tires.

PENNDOT used approximately 74,000 pounds of crumb rubber in a research project to evaluate Tyrsolv crumb rubber asphalt modifier. Costs ranged from seven to 23 percent higher than standard hot-mix asphalt. Although the Tyrsolv modifier did perform well in rut resistance, it showed no significant increase in performance over standard hot-mix asphalt. See Appendix A for additional details on the research project.

Although the cost difference was significant, PENNDOT recommended developing a provisional use specification at nine percent crumb rubber, by weight. This specification should be completed by next construction season.

PENNDOT has also approved various products containing recycled tire crumb rubber for use in road construction:

- Traffic delineation drums
- Rubberized railroad crossings
- Rubberized expansion joint-filler materials
- Noise barriers
- Manhole/inlet grade adjustment rings
- Guide rail offset blocks
- Rubber sealant to repair cracks in highways

State System of Higher Education and State Related Universities

Like the Department of Education, the State System of Higher Education (SSHE) does not itself undertake the construction of new facilities; this is done at the individual university level. However, SSHE is in a position to encourage its member universities to incorporate waste tire-derived materials in engineering and construction projects. Like Education, these would most likely consist of rubberized surfaces for athletic fields, as well as surfaces for walkways, shredded tire aggregate for landscaping purposes, rubberized surfacing for parking lots, and rubber parking bumpers. Some universities under SSHE have begun this process already; Shippensburg University has recently completely a rubberized turf soccer field. There is also the possibility for universities, like Correction facilities, to use TDF as a supplement to their fuel sources at their power generation facilities. However,

many of the existing State Related Universities facilities are being converted from coal burning to natural gas burning.

Commonwealth Recycling of Waste Tires

Pennsylvania has a tire recycling contract with three vendors. The Pennsylvania State Police and the Departments of General Services (DGS) and Corrections regularly use these vendors to recycle used tires. State colleges and universities use the vendors when they install new tires on their state vehicles. The Pennsylvania Game Commission and the Departments of Conservation and Natural Resources (some of the fish hatcheries) and Health (some of the state hospitals) have made use of the contract.

DGS Vehicle Management recently developed and implemented a more structured auto service contract, so there is reason to believe that all tires from state vehicles are being recycled. A survey, however, would be valuable.

Pennsylvania Waste Tire Program Update

Other Act 111 Accomplishments

The Waste Tire Recycling Act (Act 190 of 1996) provides for the proper disposal of waste tires and the cleanup of stockpiled tires; authorized investment tax credits for utilizing waste tires; and provided remediation grants for the cleanup of tire piles. Act 111 of 2002 added definitions; further provided for disposal of whole waste tires, for the priority enforcement list and for remediation grants; provided for remediation liens, for an authorization program, for documentation and recordkeeping, for revocation of authorization and for collection programs; authorized civil penalties; and repealed obsolete provisions relating to tire recycling investment tax credits.

Section 106.1 (Waste Tire Hauler) Authorization Program

The Department has implemented the Waste Tire Hauler Authorization program as required under the Act 111 amendments to Act 190. To date, the Department has authorized approximately 775 individuals, companies and agencies that collect and haul waste tires to processors.

A waste tire hauler who wishes to operate in Pennsylvania must obtain authorization from the Department. The authorization is not transferable. No one in Pennsylvania may accept whole used or waste tires from a waste tire hauler that does not have a valid authorization.

Each hauler pays an annual authorization fee of \$50. These fees are deposited into the Used Tire Pile Remediation Restricted Account (established in Section 110) and are used to administer the program. The Department is responsible for evaluating the fee and modifying it to cover actual program costs.

The Department maintains a registry of waste tire haulers who operate in Pennsylvania. The registry includes the hauler's name, address and telephone number, and is available through the PaPowerPort at www.state.pa.us Pa Keyword "Waste Tires" or on the PA DEP website at http://www.dep.state.pa.us/WM_APPS/TIRES/. The Department also maintains a toll-free number that anyone in Pennsylvania can call to request information from the registry. Callers also may use the toll-free number to report any violations.

Section 107 Priority Enforcement List

The Department maintains a statewide list of waste tire sites containing stockpiles of more than 10,000 waste tires (known or estimated). This list is available through the PaPowerPort at www.state.pa.us Pa Keyword "Waste Tires". The waste tire sites are ranked according to their potential for creating environmental health and safety hazards. The Department also maintains a statewide list of waste tire sites that contain 1,500 or more tires.

Pennsylvania's Priority Waste Tire List is updated monthly as tire piles are discovered and others are remediated. See Appendix C for a list of priority waste tire piles currently located in Pennsylvania.

Section 114 Waste Tire Collection Programs

Section 114 of Act 111 authorized the Department to transfer \$250,000 per fiscal year from the Recycling Fund from July 1, 2002 through the fiscal year beginning July 1, 2006 for waste tire collection programs. These collection programs help reduce risks from improper storage and disposal of tires at individual residences. As of July 1, 2004 the Department has issued \$141,000 in grants to 29 municipalities that has resulted in the collection of 112,005 tires.

Proposed Amendments to the Act

Section 104. Definitions

The Department supports proposed amendments to the definition of waste tire. This change would help prevent the speculative accumulation and improper storage of waste tires and ultimately the creation of new waste tire piles.

Section 106.1 (Waste Tire Hauler) Authorization Program

The Department supports proposed amendments to Section 106.3 (c) Waste tire registry to state that “No person may provide whole used or waste tires to a waste tire hauler that does not have a valid authorization as provided under this act.” This would close the loop on responsible parties by making generators responsible for the proper management of waste tires .

Section 111.1 Remediation Liens

The Department supports proposed legislation that improves our ability to utilize remediation liens for waste tire pile cleanups funded by the Department.

Waste Tire Pile Remediation

Since 1996, the Commonwealth has used a combination of grant programs, Department funds, and enforcement to remove approximately 23.8 million waste tires from known waste tire piles. The original funding under Act 190 ended in 2002. Additional funding as part of the Governor's budget was not provided in FY 02/03 and FY 03/04. See **Appendix B** for a list of waste tire piles cleaned up since the previous report.

Today, approximately 12.5 million waste tires remain in approximately 70 stockpiles across Pennsylvania. More than 8.5 million waste tires (68 percent) are contained in piles with no viable responsible party. The largest are the Max and Martha Starr pile in Columbia County, which contains an estimated six to ten million waste tires, and Natrl Windgap in Northampton County, with an estimated 1.2 million waste tires.

The remaining 4 million waste tires are located at commercial sites where the Department is taking enforcement and compliance actions to reduce the number of tires, and the risks from fires and mosquitoes. The largest commercial pile is Mahantango Enterprises in Juniata County with an estimated 1.5 million tires.

The Department maintains a statewide list of waste tire sites containing stockpiles of more than 10,000 waste tires (known or estimated). This list is available through the PaPowerPort at www.state.pa.us Pa Keyword "Waste Tires". The waste tire sites are ranked according to their potential for creating environmental health and safety hazards. The Department also maintains a statewide list of waste tire sites that contain 1,500 or more tires. Priority waste tire piles are piles designated by DEP in accordance with Act 190 as those that contain 10,000 or more tires, and are considered to be an environmental risk given their proximity to nearby communities. Pennsylvania's Priority Waste Tire List is updated monthly as tire piles are discovered and others are remediated. See **Appendix C** for a list of priority waste tire piles currently located in Pennsylvania.

Act 190 Waste Tire Remediation Grant Program 1997-2002

To facilitate the remediation of abandoned waste tire piles in Pennsylvania, Act 190 established the Waste Tire Pile Remediation Grant Program. The program was designed to reduce and eliminate priority waste tire piles in the Commonwealth.

Act 190 provided \$1 million annually for five years from the Recycling Fund, beginning in 1997, to fund grants awarded under the Act 190 Waste Tire Remediation Program. Act 190 Waste Tire Remediation Grants, which were awarded between 1997 and 2002, resulted in the remediation of an estimated 4.2 million waste tires from 21 priority piles scattered across the Commonwealth. This grant is no longer available.

Municipal Waste Tire Remediation Grant Program 1997-2003

In Fiscal Year 1997, the Pennsylvania General Assembly approved the allocation of additional resources to remediate priority waste tire piles located in the Commonwealth. This funding, which supplemented existing appropriations under Act 190, created the Municipal Waste Tire Remediation Grant Program. Since 1998, a total of \$14 million was allocated for grants under this program.

Under the Political Subdivision Grant Program, grants were awarded to remediate only piles containing 5,000 or more tires. Only municipalities could apply and receive funding under the Municipal Waste Tire Grant Program.

Since 1998, DEP has awarded a total of 61 Political Subdivision Grants resulting in the remediation of an estimated 8.3 million tires. The tires remediated under the program were used for TDF and engineering uses. The remaining tires were recycled and reused to construct reefs, crumb rubber for products, and engineered fill. (See **Appendix B**.)

Tire Shredder 2000-2004

In 2000, DEP contracted with the Recycling Environmental Group Inc. of Bloomsburg, Columbia County, to process and eliminate waste tire piles with an on-site mobile tire shredder. The shredder, which was transported to waste tire piles across the state, processed waste tires into tire chips. The chips were transported directly to end users and used in a variety of applications, including landfill construction, other civil engineering uses and TDF.

Since 2000, DEP has remediated an estimated 733,943 waste tires using the waste tire shredder at a cost of \$1 million from the Solid Waste Abatement Fund.

Waste Tire Collection Programs 2002-2007

Section 114 of Act 111 authorized the Department to transfer \$250,000 per fiscal year from the Recycling Fund from July 1, 2002 through the fiscal year beginning July 1, 2006 for waste tire collection programs. These collection programs help reduce risks from improper storage and disposal of tires at individual residences. As of July 1, 2004 the Department has issued \$141,000 in grants to 29 municipalities that has resulted in the collection of 112,005 tires.

Forest Lands Beautification Program

The Department of Conservation and Natural Resource, through its Forest Lands Beautification Program, has been identifying and cleaning up illegal dumpsites. To date, more than 220 of almost 300 illegal dumpsites have been cleaned through the efforts of contractors and more than 2,000 volunteers. These cleanups have resulted in the removal and proper disposal of almost 2,800 tons of household trash, more than 23,000 tires, 280 tons of scrap metal and 640 tons of concrete.

Enforcement and Compliance

The DEP has used its authority under Act 190, as amended by Act 111, to develop regulations to require tire pile owners to reduce the risks associated with waste tire piles and cleanup waste tires stored longer than one year and waste tire piles greater than five acres. Some of the remediations conducted were accomplished as a result of regulatory actions by DEP, including consent orders and agreements (CO&A). Under these agreements, the waste tire cleanup is paid for by the owner of the pile in lieu of paying a fine or penalty to DEP. In other situations, cleanups were accomplished by DEP through the use of Hazardous Sites Cleanup Act (HSCA) funds and through the voluntary actions of pile owners.

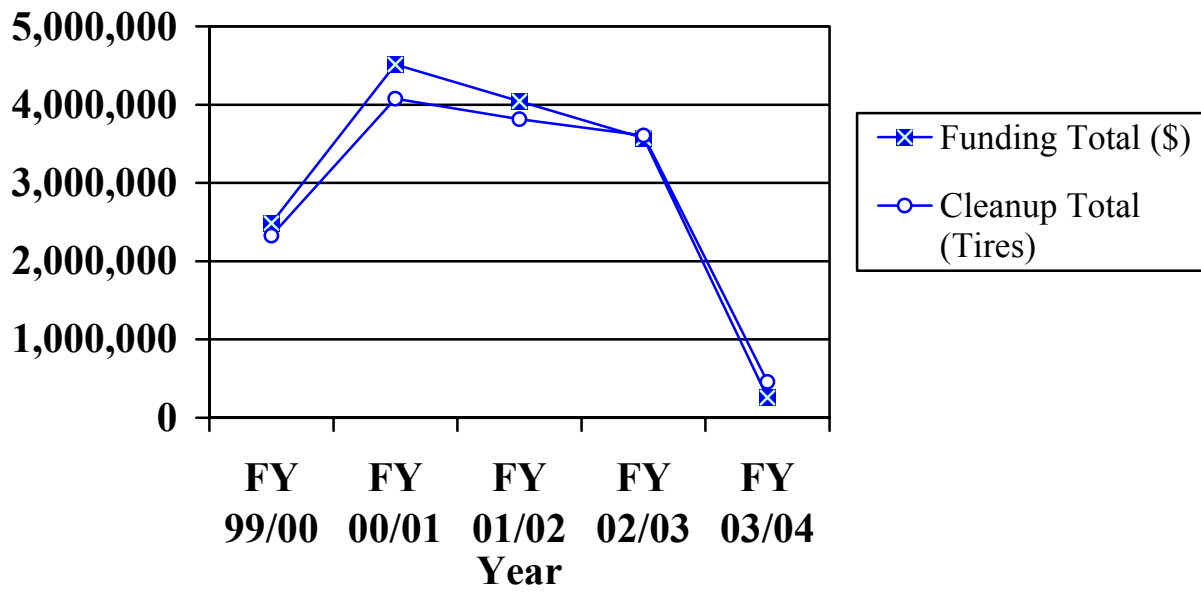
DEP's regulatory and compliance programs continue to be a key factor in Pennsylvania's overall waste tire recycling success. Nearly 13.5 million, or over 77%, of the waste tires at commercial sites have been remediated as a direct result of regulatory and compliance actions taken by DEP. DEP's enforcement and compliance initiatives make an effective strategy to eliminate and prevent waste tire stockpiling in the Commonwealth.

Together, DEP's cleanup programs and compliance initiatives make an effective strategy to eliminate and prevent the stockpiling of waste tires.

Funding	FY99/00	FY00/01	FY01/02	FY02/03	FY03/04
Act 190	\$453,471	\$1,417,721	\$1,000,000	\$0	\$0.00
General Fund	\$1,651,579	\$1,611,446	\$1,840,892	\$1,387,005	\$0.00
Other	\$379,957	\$1,484,603	\$1,204,426	\$2,184,911	\$255,729
Total	\$2,485,007	\$4,513,768	\$4,045,318	\$3,571,916	\$255,729

Results	FY99/00	FY00/01	FY01/02	FY02/03	FY03/04
Act 190	325,804	1,175,046	638,829	0	0
General Fund	1,351,712	1,188,660	1,470,805	1,067,214	0
Other	363,700	1,406,430	912,502	1,763,651	164,114
Enforcement	286,800	304,200	791,311	775,921	292,229
Total	2,228,016	4,074,336	3,813,447	3,606,786	456,343

Summary of Funding and Results



Future Challenges

Today, through the implementation of Act 190 and other related waste tire recycling programs, more Pennsylvania tires are being beneficially used than generated. This trend is not only prevalent in Pennsylvania, but is true across the nation. According to the EPA Region 5 publication on 'Scrap Tires in the Great Lakes Region', the United States now recycles about 78 percent of the 281 million waste tires generated each year.

The beneficial uses for waste tires include tire derived fuel (TDF), engineering applications, crumb rubber, retreads, and export.

The majority of waste tires are burned as TDF in cement kilns, lime kilns, and waste-to-energy facilities. TDF is also used in large industrial furnaces. The usage of waste tires in kilns and industrial furnaces provides both an **income** and energy source. The number and location of facilities capable of burning waste tires limits the usage of TDF. The price of other energy sources also influences the usage of waste tires. Until waste tires have a positive value, it is expected that TDF will continue to be one of the most significant uses for waste tires.

The Journal for Municipal Solid Waste Professionals reported a 250% increase in the number waste tires used in civil engineering applications from 1994 to 2000. Examples of civil engineering applications for waste tires include artificial reefs, breakwaters, backfill for wall and bridge abutments, highway crash barriers, landfill construction, levee reinforcement, slope stabilization, erosion control, sub-grade fill, and embankments. Most of these construction and engineering activities use waste tires as a substitute for traditional fill or drainage materials. The usage of waste tires in construction activities is influenced by availability, economic and technical constraints.

The market for crumb rubber from waste tires continues to grow. New market entrants, failed enterprises, imports, overcapacity, and weak demand for high volume uses of crumb rubber such as asphalt have negatively influenced the crumb rubber market. Examples of uses for crumb rubber include animal bedding, automotive parts, equestrian riding areas, filter media, molded and extruded products, mulch, noise barriers, parking lot surfaces, railroad crossings, rubber matting, rubberized asphalt, soil amendments and traffic cones.

The retread market for passenger car tires has decreased dramatically due to the low cost of new tires and market acceptance of retread tires for passenger cars. The retread market for commercial truck tires continues to be strong due to the high cost of new tires and wide-spread industry use of retreads.

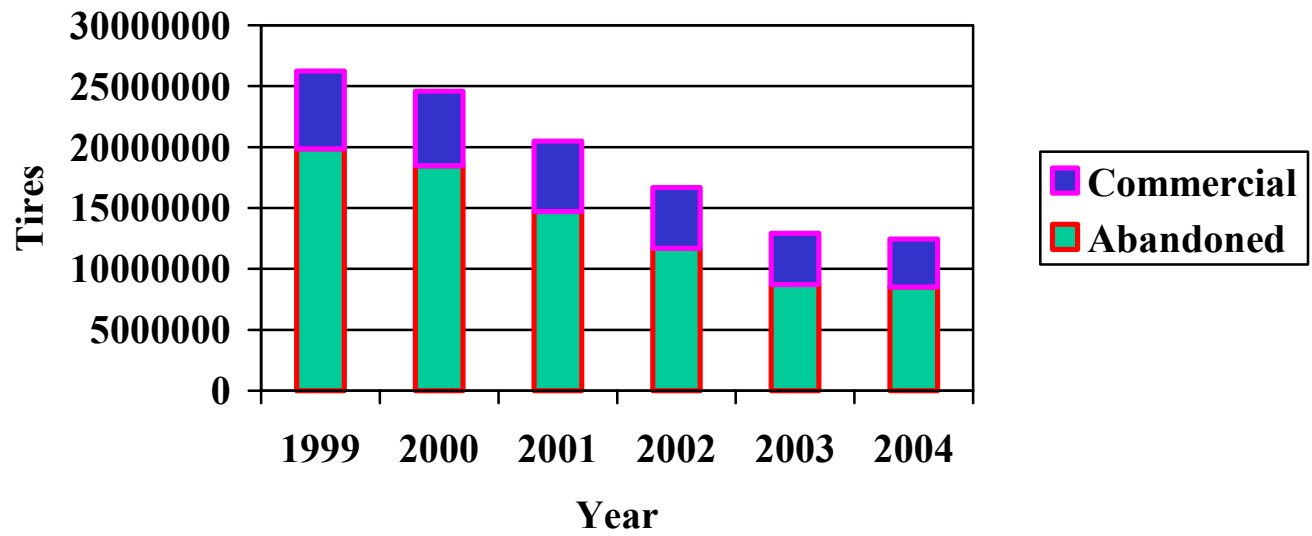
Bottom Line

While beneficial uses exist for waste tires, these are driven by government statutes prohibiting disposal of whole waste tires in municipal waste landfills. The **bottom line** is that waste tires remain a **financial liability** and the potential for improper management and disposal still exists. As we continue our efforts to cleanup the remaining waste tire piles we must apply a constant vigilance on waste tire storage and processing facilities to prevent the creation of new tire piles.

The Department will continue to:

- Support amendments to the Waste Tire Act to improve our ability to prevent generators from using unauthorized transporters and enable the Department to place liens on waste tires sites cleaned up by the Commonwealth.
- Support waste tire markets by promoting construction, engineering, and product uses for waste tires;
- Encourage DGS, PENNDOT, and other state agencies to use waste tires and products produced from waste tires;
- Reduce the risks at commercial waste tire piles using enforcement and compliance;
- Cleanup the remaining 8.5 to 12.5 million waste tires at abandoned waste tire piles; and
- Seek cost recovery and penalties from responsible parties and property owners for state funded cleanups.

Tire Pile Inventory



Appendix A

Summary of PENNDOT Evaluations of Recycled Tire Crumb Rubber in Hot-mix Asphalt (HMA) Pavement Courses

SR Section	County	Rubber Process	Year Built	Job Length (miles)	Quantity of HMA (tons)	Quantity of Rubber Used (lbs.)	Cost per Ton	Performance Comments
0819	Fayette	Wet	1991	2.0	2,213	27,441	\$44.91 (Control = Std. ID-2 Wearing = \$28.06) 60% higher than control	ID-2 Wearing with Crumb Rubber exhibited longitudinal and transverse cracking beginning two years after construction. By the third year, there was a significant difference in the amount of cracking between the crumb rubber section and the control section. By the fourth year, PENNDOT Maintenance had to seal the cracks in the crumb rubber section, resulting in additional maintenance costs.
0041-0010	Chester	Wet	1996	0.3	400	Unknown	\$32.10 7% higher than control	During production of the hot-mix asphalt with rubber, the crumb-rubber-modified (CRM) liquid asphalt binder was pumped into a nonagitated storage tank in the hot-mix plant. The rubber in the liquid asphalt binder separated and floated to the top of the tank. The amount of rubber actually incorporated into the final hot-mix asphalt is unknown.
0070-009	Fulton	Wet	1998	9.9	3,900	35,800	\$48.95 23% higher than control	Superpave 12.5 mm Wearing Course with Rubber. Some cracking in the rubber section. Rut performance is equal to control section (Superpave 12.5 mm Wearing Course, PG 64-23, stretched 1 year before rubber section).
0081-050	Schuylkill	Dry	1998	5.17	3,689	38,292	\$78.00 13% higher than control	Superpave 19mm Wearing Course with Rubber. Rubber section rut performance is approx. equal to control section (Superpave 19mm Wearing Course, PG 76-22).
1004	Union	Dry	2001	0.11	39	1,045	Crumb rubber mixture provided by producer at same price as control as a test. Estimated cost increase per ton for crumb rubber mixture (material only) is \$3.50/ton (approx. 10% higher than control).	Crumb rubber was added to a fine-grade ID-2 Wearing Course and compared to a fine-grade ID-2 Wearing Course without crumb rubber (control). To date, field performance is comparable to the control. Additional Superpave Performance Tests were conducted on the mixtures collected from the field. Test results indicated crumb rubber mixture should be used with caution for rut resistance when compared to the control. Crumb rubber mixture has similar fatigue and low temperature properties as control.

APPENDIX B

Waste Tire Cleanup Projects

County	Municipality	Site	Contractor/Grantee	Fund	Amount	Tires	Year
Allegheny	Mckees Rocks	Pittsburgh Limosine	NA ¹	NA	NA	1,500	2003
Allegheny	Moon	Sewickly Bridge Barges	NA	NA	NA	400,000	2003
Armstrong	Cowanshannock	Hetrick	Recycling Envir. Group	Other	\$21,818	13,952	FY03/04
Armstrong	Kittanning	Cochran	Kittanning Township	Gen. Fund	\$117,492	175,461	FY01/02
Armstrong	Kittanning	Cochran	Kittanning Township	Other	\$68,649	135,326	FY01/02
Armstrong	Plum Creek	Kaufman	Plumcreek Township	Gen. Fund	\$11,880	30,000	FY01/02
Armstrong	Plum Creek	Kaufman	Plumcreek Township	Other	\$25,000	30,538	FY01/02
Armstrong	Plum Creek	Kaufman	Plumcreek Township	Other	\$12,953	9,666	FY01/02
Armstrong	Plum Creek	Kaufman	Plumcreek Township	Gen. Fund	\$32,632	24,352	FY02/03
Armstrong	Wayne	Snyder	Wayne Township	Gen. Fund	\$34,230	19,863	FY00/01
Bedford	E. Providence Twp	Griffin 1	Carbon Services Corp.	Act 190	\$16,784	7,813	FY01/02
Bedford	E. Providence Twp	Griffin 2	Carbon Services Corp.	Act 190	\$140,037	86,877	FY01/02
Berks	Amity Twp.	TPS	Recycling Envir. Group	Act 190	\$375,057	230,804	FY00/01
Berks	Amity Twp.	TPS	Recycling Envir. Group	Other	\$107,498	71,533	FY00/01
Berks	Amity Twp.	TPS	Recycling Envir. Group	Other	\$139,378	97,292	FY01/02
Berks	Amity Twp.	TPS	Recycling Envir. Group	Other	\$139,378	97,292	FY02/03
Berks	Bern, Centre, Penn	Jungle Road	Recycling Envir. Group	Other	\$186,040	125,136	FY01/02
Berks	Bern, Centre, Penn	Jungle Road	Recycling Envir. Group	Other	\$186,040	125,136	FY02/03
Berks	Sinking Springs	Norfolk Southern	NA	NA	NA	1,500	2004
Berks	Tulpehocken Twp.	Stone Road	Recycling Envir. Group	Other	\$148,914	100,883	FY01/02
Berks	Tulpehocken Twp.	Stone Road	Recycling Envir. Group	Other	\$148,914	100,883	FY02/03
Blair	Antis	Burch	Antis Township	Other	\$8,895	4,471	FY02/03
Blair	Antis	Forshey's Garage	NA	NA	NA	2,000	2003
Blair	Blair	Walter's Auto Wrecking	NA	NA	NA	600	2003
Blair	Holidaysburg	Holiday Tire Service	NA	NA	NA	1,000	2003
Blair	Snyder Twp.	S & R Recycling	Carbon Services Corp.	Act 190	\$492,145	578,994	FY00/01
Blair	Snyder Twp.	S & R Recycling	Carbon Services Corp.	Other	\$550,000	647,059	FY00/01
Bradford	Canton	Stev Jennings Farm	Leroy Township	Gen. Fund	\$25,032	20,667	FY01/02
Bucks	Fairmount Twp	Houssock	Carbon Services Corp.	Other	\$109,355	68,238	FY01/02
Bucks	Falls Twp.	Huston Junkyard	NA	NA	NA	25,000	2004
Bucks	Falls Twp.	Huston Junkyard	Magnum Industries, Inc.	Act 190	\$70,000	53,252	FY00/01
Bucks	Falls Twp.	Huston Junkyard	Magnum Industries, Inc.	Act 190	\$38,350	29,175	FY01/02
Bucks	Haycock Twp.	Calabrette	Recycling Envir. Group	Act 190	\$34,650	21,323	FY00/01
Bucks	Springfield Twp.	Gemstar	Springfield Twp.	Other	\$342,377	327,728	FY00/01
Bucks	Springfield Twp.	GemStar	Recycling Envir. Group	Other	\$21,832	0	FY01/02
Bucks	Springfield Twp.	GemStar	Recycling Envir. Group	Other	\$21,832	0	FY02/03
Butler	Cherry	Meyle Property	NA	NA	NA	5,000	2003
Butler	Summit Twp.	Frank Miller	Summit Twp.	Gen. Fund	\$400,000	261,445	FY00/01
Cambria	Adams Twp.	Kenneth Krestar	Adams Township	Gen. Fund	\$5,834	4,522	FY01/02

¹ NA = Not applicable (Tires were cleaned up by responsible party).

County	Municipality	Site	Contractor/Grantee	Fund	Amount	Tires	Year
Cambria	Adams Twp.	Kenneth Krestar	Adams Township	Other	\$56,069	44,748	FY01/02
Cambria	Clearfield Twp.	Balzano/Cree	Clearfield Township	Gen. Fund	\$108,000	99,815	FY02/03
Cambria	Clearfield Twp.	Balzano/Cree	Clearfield Township	Other	\$55,077	50,812	FY02/03
Cambria	Portage Twp.	Dandrea	Portage Township	Other	\$131,681	120,028	FY02/03
Cambria	Richland Twp.	Teno's Junkyard	Richland Township	Other	\$54,972	38,829	FY02/03
Cambria	Susquehanna	Allegheny Env.	Recycling Envir. Group	Other	\$36,262	23,532	FY03/04
Centre	Bellefonte	Wolf Tire Pile	Centre County	Gen. Fund	\$17,750	13,867	FY01/02
Centre	College	Steward Tire Pile	NA	NA	NA	30,000	2001
Centre	Snow Shoe	Potisnock Tire Pile	NA	NA	NA	104,000	2002
Chester	E. Fallowfield Twp	Prangs	Magnum Industries, Inc.	Act 190	\$32,421	18,013	FY00/01
Clarion	Knox Borough	Harris Truck Company	NA	NA	NA	1,500	2003
Clarion	Madison Twp.	Tire Tread Shredder	Carbon Services Corp.	Act 190	\$416,378	259,038	FY01/02
Clearfield	Woodward	Natallies Tire Pile	NA	NA	NA	12,000	2001
Clearfield	Woodward Twp	Natalie Property	Clearfield County	Gen. Fund	\$13,961	13,296	FY00/01
Clinton	Bald Eagle	Melvin Smith	Recycling Envir. Group	Other	\$35,811	22,695	FY03/04
Columbia	Greenwood	Greg Starr Tire Pile	NA	NA	NA	41,129	2004
Columbia	Greenwood	Karns Tire Pile	NA	NA	NA	17,400	2003
Columbia	Madison Twp.	Lolota site	Recycling Envir. Group	Act 190	\$70,000	50,007	FY00/01
Crawford	Cambridge	Cambridge Springs Tire	NA	NA	NA	10,000	2002
Crawford	Hartstown	Mishorich	West Fallowfield Twp.	Other	\$128,657	79,585	FY01/02
Crawford	Rome	Robert Hopkins Property	NA	NA	NA	2,000	2003
Crawford	Vernon	Bob Gawne Property	NA	NA	NA	2,300	2003
Dauphin	Wiconiosco	One Stop Recycling	NA	NA	NA	8,000	2000
Erie	Conneaut	Parknow	Recycling Envir. Group	Other	\$298,110	212,936	FY00/01
Erie	Conneaut	Parknow	Erie County	Gen. Fund	\$40,935	38,025	FY02/03
Erie	Conneaut	Parknow	Erie County	Other	\$37,935	38,025	FY02/03
Erie	Conneaut	Parknow	Erie County	Other	\$37,935	38,025	FY02/03
Erie	Elk Creek	Kaufmann	Erie County	Gen. Fund	\$3,000		FY02/03
Erie	Elk Creek	Kaufmann	Erie County	Other	\$42,122	42,222	FY02/03
Erie	Elk Creek	Kaufmann	Erie County	Other	\$42,122	42,223	FY02/03
Erie	Erie	Cohen	City of Erie	Other	\$56,088	57,673	FY02/03
Erie	Erie Co.	Lenhart	Erie County	Gen. Fund	\$2,477		FY02/03
Erie	Erie Co.	Lenhart	Erie County	Other	\$46,080	45,190	FY02/03
Erie	Erie Co.	Lenhart	Erie County	Other	\$46,080	45,190	FY02/03
Erie	Fairview	John Hook	Recycling Envir. Group	Act 190	\$160,569	114,692	FY00/01
Erie	West Springfield	Bihler Tire Site	Springfield Township	Other	\$22,235	11,827	FY01/02
Fayette	Georges Twp.	Georges-Fairchance Rd	Georges Township	Other	\$32,225	17,139	FY02/03
Fayette	Menallen Twp.	Kushak	Carbon Services Corp.	Act 190	\$120,038	81,003	FY01/02
Fayette	Smithfield	Bankhead-Humbert	Springhill Township	Gen. Fund	\$146,675	188,771	FY00/01
Franklin	Montgomery	Wishards Tire Processing	NA	NA	NA	50,000	2001
Greene	Jefferson Twp.	Ruschell	Greene County	Other	\$18,553	14,606	FY02/03
Greene	Mt. Morris	Kiger	Perry Township	Gen. Fund	\$225,109	127,023	FY00/01
Greene	Mt. Morris	Kiger	Perry Township	Act 190	\$182,879	107,961	FY00/01
Greene	Mt. Morris	Kiger	Perry Township	Gen. Fund	\$400,000	226,019	FY01/02
Greene	Mt. Morris	Kiger	Perry Township	Gen. Fund	\$400,000	225,708	FY02/03
Jefferson	Knox Township	Rhodes Salvage	Jefferson County	Gen. Fund	\$104,953	87,827	FY00/01

County	Municipality	Site	Contractor/Grantee	Fund	Amount	Tires	Year
Juniata	Milford	Juniata Tire Sales	NA	NA	NA	6,000	2004
Lancaster	Earl	New Holland	NA	NA	NA	10,000	2004
Lancaster	East Hempfield	Paquins Tire Services	NA	NA	NA	5,000	2004
Lancaster	Fulton Twp.	McFadden Dump	Fulton Township	Other	\$399,894	403,493	FY02/03
Lancaster	W Donegal	Mease's Auto Salvage	NA	NA	NA	15,000	2004
Lawrence	New Castle	Hovis Auto	NA	NA	NA	5,000	2002
Lawrence	New Castle	Leonard Property	Lawrence County	Gen. Fund	\$38,750	25,000	FY01/02
Lawrence	New Castle	Leonard Property	Lawrence County	Other	\$33,499	21,612	FY01/02
Lawrence	New Castle	Martha Property	Lawrence County	Gen. Fund	\$14,958	13,080	FY01/02
Lawrence	New Castle	Martha Property	Lawrence County	Other	\$38,847	33,989	FY01/02
Lawrence	Shenango	New Castle Auto Wrecking	NA	NA	NA	7,000	2002
Lawrence	Slippery Rock	McCormick Auto Sales	NA	NA	NA	5,000	2002
Lawrence	Slippery Rock Twp	Jim Frew Property	Lawrence County	Gen. Fund	\$13,152	8,485	FY00/01
Lebanon	East Hanover	Mike Geesaman	NA	NA	NA	70,000	2004
Lebanon	Heidelberg	Keller Brothers	NA	NA	NA	2,000	2004
Luzerne	Bear Creek	Rec. Realty	Recycling Envir. Group	Other	\$31,305	24,081	FY00/01
Luzerne	Butler	New Land Dev. Co.	NA	NA	NA	300	2003
Luzerne	Duryea	Coxton Tire Pile	Duryea Borough	Gen. Fund	\$136,312	99,906	FY00/01
Luzerne	Duryea	Coxton Tire pile	Duryea Borough	Gen. Fund	\$400,000	293,080	FY01/02
Luzerne	Duryea	Coxton Tire Pile	Duryea Borough	Gen. Fund	\$400,000	293,169	FY02/03
Luzerne	Edwardsville	Broody	Edwardsville Borough	Other	\$122,202	98,431	FY02/03
Luzerne	Fairmount Twp.	Houssock	Carbon Services Corp.	Act 190	\$186,108	115,863	FY01/02
Luzerne	Foster	Beltrami Enterprises Inc.	NA	NA	NA	70,000	2001
Luzerne	Hazel Twp.	Bonner	Carbon Services Corp.	Act 190	\$82,304	59,060	FY01/02
Luzerne	Hazleton City	Hazleton Scrap Recycling	NA	NA	NA	10,000	2002
Luzerne	Jackson	Donald Malak	NA	NA	NA	5,000	2003
Luzerne	Jackson	R. Malak Scrapyard	NA	NA	NA	8,000	2003
Luzerne	Larksville Boro.	Wesley Cooper	Recycling Envir. Group	Other	\$37,544	28,880	FY00/01
Luzerne	Sugarloaf	G&R Zanolini Auto	NA	NA	NA	500	2003
Luzerne		Luzerne County	Recycling Envir. Group	Other	\$133,828	77,057	FY01/02
Luzerne		Luzerne County	Recycling Envir. Group	Other	\$133,828	77,057	FY02/03
Lycoming	Williamsport	Lycoming Supply Tire Pile	NA	NA	NA	9,811	2002
Lycoming	Williamsport	Fagano's	Recycling Envir. Group	Other	\$68,211	43,640	FY03/04
Lycoming	Williamsport	Mundies	NA	NA	NA	5,000	2002
McKean	Bradford Twp.	Kohler	McKean County	Gen. Fund	\$76,318	33,978	FY00/01
McKean	Foster	Tuna Cross Road	Foster Township	Other	\$17,116	22,977	FY02/03
McKean	Fulton Twp.	Mace Recycling	Recycling Envir. Group	Other	\$223,940	201,799	FY02/03
McKean	Fulton Twp.	Mace Recycling	Recycling Envir. Group	Other	\$93,628	60,295	FY03/04
Mercer	Shenango	Rt 18 Auto Wrecking	NA	NA	NA	5,000	2003
Mercer	West Salem	Fenton Auto Shop	NA	NA	NA	2,000	2003
Mifflin	Armagh	Aumiller's Autowreckers	NA	NA	NA	7,000	2003
Monroe	Long Pond	David Smith	Tunkhannock Township	Gen. Fund	\$13,907	10,781	FY00/01
Monroe	Stroud wp.	Jahada	Stroud Township	Other	\$49,278	31,147	FY02/03
Montgomery	Lower Providence	Fairview Motors	NA	NA	NA	50,000	2001
Montgomery	Pottstown	Pete Peck	NA	NA	NA	45,000	2003
Northampton	Wind Gap	Natri Windgap	Wind Gap Borough	Gen. Fund	\$399,950	420,729	FY01/02

County	Municipality	Site	Contractor/Grantee	Fund	Amount	Tires	Year
Northampton	Wind Gap	Natrl Windgap	Wind Gap Borough	Gen. Fund	\$399,962	386,145	FY02/03
Northumberland	Mt. Carmel	Boylan's Garage	NA	NA	NA	19,200	2001
Northumberland	Mt. Carmel	Madonna Tire Pile	NA	NA	NA	1,500	2003
Northumberland	Upper Augusta	Hovenstine Tire Pile	NA	NA	NA	3,500	2002
Northumberland	Upper Augusta	Shipman Tire Pile	NA	NA	NA	10,000	2003
Perry	Marysville	Weigle	Rye Township	Gen. Fund	\$399,978	288,902	FY00/01
Perry	Marysville	Weigle	Rye Township	Gen. Fund	\$400,000	239,880	FY01/02
Perry	Marysville	Weigle	Rye Township	Other	\$30,248	18,140	FY01/02
Perry	Penn	Conrail/Norfolk Southern	NA	NA	NA	2,000	2001
Philadelphia	Philadelphia	All State Tire	NA	NA	NA	300,000	2002
Schuylkill	Mahanoy	RNH Recovery	NA	NA	NA	14,000	2001
Snyder	Perry	Pyle Tire Pile	NA	NA	NA	7,000	2004
Somerset	Boswell	Gindlesperger	Jenner Township	Gen. Fund	\$24,960	28,526	FY00/01
Somerset	Conemaugh	FHA	NA	NA	NA	2,000	2002
Somerset	Paint Twp.	McCreary Pile	Paint Township	Other	\$30,349	22,481	FY02/03
Somerset	Windber	Robert Hostensky	NA	NA	NA	3,000	2002
Susquehanna	Forest City Boro	Kost Tire	NA	NA	NA	325,000	2002
Susquehanna	Great Bend	Intst Bag&Burlap	Great Bend Township	Other	\$28,085	18,849	FY01/02
Susquehanna	Hallstead Boro	Singer Tire	NA	NA	NA	7,500	2003
Susquehanna	Montrose	Duane Brunges	Bridgewater Twp.	Gen. Fund	\$1,204	860	FY00/01
Susquehanna	New Milford	Harold Dibble	New Milford Twp.	Other	\$3,075	2,078	FY01/02
Tioga	Covington	Travis Thompson	NA	NA	NA	7,621	2003
Tioga	Hamilton	John Stempeck	NA	NA	NA	9,600	2004
Tioga	Liberty	Matt's Auto Salvage	NA	NA	NA	6,000	2000
Tioga	Millerton	Ernest Wright	Jackson Township	Gen. Fund	\$9,246	8,500	FY01/02
Tioga	Millerton	Ernest Wright	Jackson Township	Other	\$1,272	1,169	FY01/02
Tioga	Richmond	Costy's Junk Yard	NA	NA	NA	32,000	2001
Tioga	Richmond	Robinson's Salvage Yard	NA	NA	NA	4,800	2000
Tioga	Rutland	Paul Wood Tire Pile	NA	NA	NA	18,000	2000
Union	West Buffalo	Duke Property	Union County	Other	\$33,182	21,439	FY02/03
Venango	Franklin	Barbers Bros	Venango County	Other	\$16,489	36,369	FY01/02
Venango	Okland	Gifford Rd Tire Pile	NA	NA	NA	10,000	2001
Warren	Brokenstraw	West End Auto	NA	NA	NA	13,000	2003
Warren	Pine Grove	Norths Auto Salvage	NA	NA	NA	20,000	2003
Wayne	Berlin	Alex Barletta	NA	NA	NA	200	2003
Wayne	Palmyer	Teeples Auto Salvage	NA	NA	NA	100,000	2004
Westmoreland	Ligonier	Weimer Farm	Ligonier Township	Gen. Fund	\$20,686	18,997	FY00/01
Wyoming	Exeter	Kenneth Gacek	Recycling Envir. Group	Other	\$69,193	37,083	FY02/03
Wyoming	Northmoreland	Eggleston	Recycling Envir. Group	Other	\$60,158	48,126	FY00/01
Wyoming	Northmoreland	Eggleston/Matusavige	NA	NA	NA	150,000	2000
Wyoming	Tunkhannock	Minor Burgess	Recycling Envir. Group	Other	\$57,609	46,087	FY00/01
York	Dover Township	Ronald McClintock Estate	NA	NA	NA	10,000	2003
York	Hopewell	Wilson Tire Center	NA	NA	NA	15,000	2001
York	Newberry	Reeser Tire Pile	NA	NA	NA	2,000	2002
York	Windsor	Richard Blumenthal Site	NA	NA	NA	200,000	2003
York	Windsor	Rt 124 Auto Parts	NA	NA	NA	100,000	2000

APPENDIX C

Pennsylvania Priority Waste Tire Pile List

The number of tires contained in each pile is an estimate and should not be construed as being an exact count.

A — Abandoned
C — Commercial
R — Responsible Party

July 30, 2004

County	Township	Region	Status	Pile	Size
<i>Allegheny</i>					
	Forward	Pittsburgh	R	Suchko	290,000
<i>Berks</i>					
	Long Swamp	Harrisburg	C	Krauss	30,000
<i>Bucks</i>					
	Springfield/Richland	Conshohocken	C	Moyer	20,000
<i>Butler</i>					
	Allegheny	Meadville	A	Garvin Pile	15,000
	Summit	Meadville	A	Frank Miller #2	15,000
<i>Cambria</i>					
	Dean	Pittsburgh	A	Hatch	15,000
	Gallitzin	Pittsburgh	A	Kolak	10,000
<i>Centre</i>					
	Snowshoe	Williamsport	R	Don Johnson	50,000
<i>Clearfield</i>					
	Graham	Williamsport	C	Segers Central PA Tire Pile	14,000
<i>Clinton</i>					
	Bald Eagle	Williamsport	C	Martin Motors	70,000
	Bald Eagle	Williamsport	A	Melvin Smith	16,000
<i>Columbia</i>					
	Fishing Creek	Williamsport	C	Paragon Produce Corp.	10,000
	Greenwood	Williamsport		Max & Martha Starr	5,892,000
	Madison	Williamsport	C	Swartz Salvage Yard	35,000
<i>Crawford</i>					
	Beaver	Meadville	A	Atrozskin	10,500
	Hayfield	Meadville	C	Hayes Junkyard	10,000
	Athens/Rome	Meadville	A	Process Pile	10,000
	Wayne	Meadville	C	Patterson's Auto Wrecking, Inc.	150,000
<i>Cumberland</i>					
	Lower Franklin	Harrisburg	A	Boyer	25,000
<i>Dauphin</i>					
	South Hampton	Harrisburg	C	Handiwork Contractors	50,000
<i>Fayette</i>					
	Franklin	Pittsburgh	A	Walcz Farm	15,000
	North Union	Pittsburgh	C	Dandrea Brothers Salvage	50,000

County	Township	Region	Status	Pile	Size
Franklin					
	Hamilton/Greene	Harrisburg	R	Myron Young	500,000
Greene					
	Perry	Pittsburgh	A	Kiger	100,000
	Rich Hill	Pittsburgh	A	Ray McClellan Junk Yard	100,000
Huntingdon					
	Cromwell	Harrisburg	R	Gerald Booher	200,000
Indiana					
	Armstrong	Pittsburgh	R	Mowry	28,000
	Center	Pittsburgh		Heberling	30,000
	Center	Pittsburgh		Weyandt	15,000
Juniata					
	Susquehanna	Harrisburg	C	Mahantango Enterprises	1,500,000
Lackawanna					
	Archbald Boro	Wilkes-Barre	C	Highway Auto Parts	50,000
Lancaster					
	Clay	Harrisburg	C	Larry Gehr	25,000
	Salisbury	Harrisburg	C	Earnest Barkmon	150,000
	W. Donegal	Harrisburg	C	Mease's Auto Salvage	15,000
	Bart	Harrisburg	A	John Smuckers	25,000
	Fulton	Harrisburg	A	McFadden Dump (Property)	500,000
Lawrence					
	New Castle	Meadville	C	C & V Salvage	32,000
Lebanon					
	East Hanover	Harrisburg	C	Mike Geesaman	30,000
Luzerne					
	Duryea Boro	Wilkes-Barre	A	Coxton/Scran	100,000
	Hunlock	Wilkes-Barre	C	ABC Auto Parts	10,000
	Lake	Wilkes-Barre	C	Lamoreaux	70,000
	Salem	Wilkes-Barre	C	Ed Spencers Auto Parts	12,000
Mercer					
	East Lackawannock	Meadville	C	Robert Bagnall	75,000
	Mercer Boro	Meadville	C	Mercer Auto Wreckers	12,000
	Sandy Creek	Meadville	C	Snowfields Junkyard	35,000
	West Salem	Meadville	C	Russell Fenton	35,000
	Salem	Meadville	A	Betty Anderson	12,000
Monroe					
	Hamilton	Wilkes-Barre	C	Hopkins Scrapyard	10,000
Northampton					
	Wind Gap Boro	Wilkes-Barre	A	Natr'l Windgap	1,200,000
	Stockertown	Wilkes-Barre	C	Einfalt's Salvage Yard	10,000
Perry					
	Carroll	Harrisburg	C	Perry County Metals	15,000
	Tuscarora	Harrisburg		Barry Mather	10,000

County	Township	Region	Status	Pile	Size
<i>Philadelphia</i>					
	Philadelphia	Norristown	C	DEBO	250,000
<i>Schuylkill</i>					
	North Manheim	Wilkes-Barre	C	Paul Wellers Tire Shop	10,600
<i>Snyder</i>					
	Perry	Williamsport	C	Pyle Tire Pile	50,000
	West Perry	Williamsport	C	Evendale Tires	15,000
<i>Susquehanna</i>					
	Forest Lake	Wilkes-Barre	A	Otis Graham	11,500
<i>Venango</i>					
	Sugar Creek	Meadville	C	Lowery Auto Wrecking	30,000
	President	Meadville	A	Jerry Richards	20,000
<i>Warren</i>					
	Freehold	Meadville	C	Seamens Auto Wrecking & Salvage	50,000
<i>Wayne</i>					
	Palmyra	Wilkes-Barre	C	Teeple's Auto Salvage	150,000
	Texas	Wilkes-Barre	C	Wickles Auto Salvage	10,000
<i>Westmoreland</i>					
	Murrysville	Pittsburgh	C	Export Tire Co.	30,000
<i>Wyoming</i>					
	Lemon	Wilkes-Barre	A	Martin Harvey	10,000
	Noxen	Wilkes-Barra	C	John Simoon	20,000
<i>York</i>					
	Chanceford	Harrisburg	A	Glenn Cunningham Property	50,000
	Washington	Harrisburg	A	Pine Woods	10,000

For more information, visit DEP's website at www.dep.state.pa.us , Keyword: "DEP Waste Tires."